

A small W.O.P.R., built around real boards

Gray case. Flush white lettering. Eight side-light modules. One inset touchscreen.

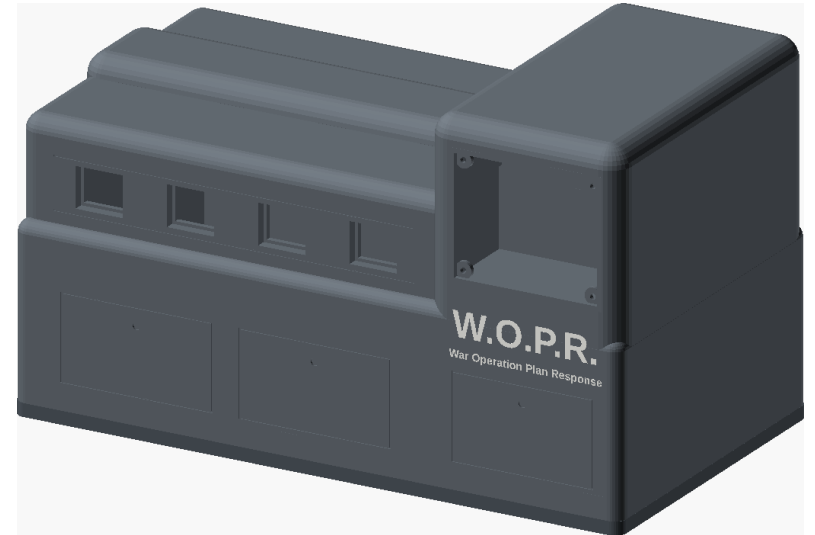
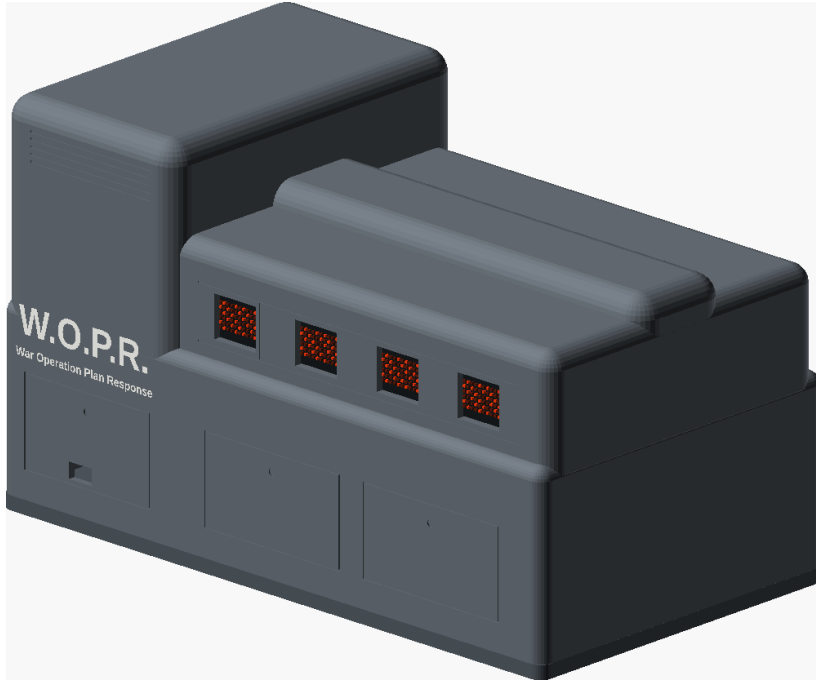


279.4 x 155 x 165 mm | 11.00 x 6.10 x 6.50 in | 3 mm nominal shell

OpenSCAD model render. Colored LEDs and display are hardware previews, not printed material.

The same design, around the back

Eight boards total: four on each long side. The small LED clusters are shown at their real size.



CASE WITHOUT ELECTRONICS

The silhouette follows the low central case, rounded shoulders, raised spine and tall end tower.

Both long sides carry white W.O.P.R. / War Operation Plan Response lettering.

The main touchscreen is on one side, directly above its logo.

Four separate clusters light up on each side; there are 480 LEDs across all eight boards.

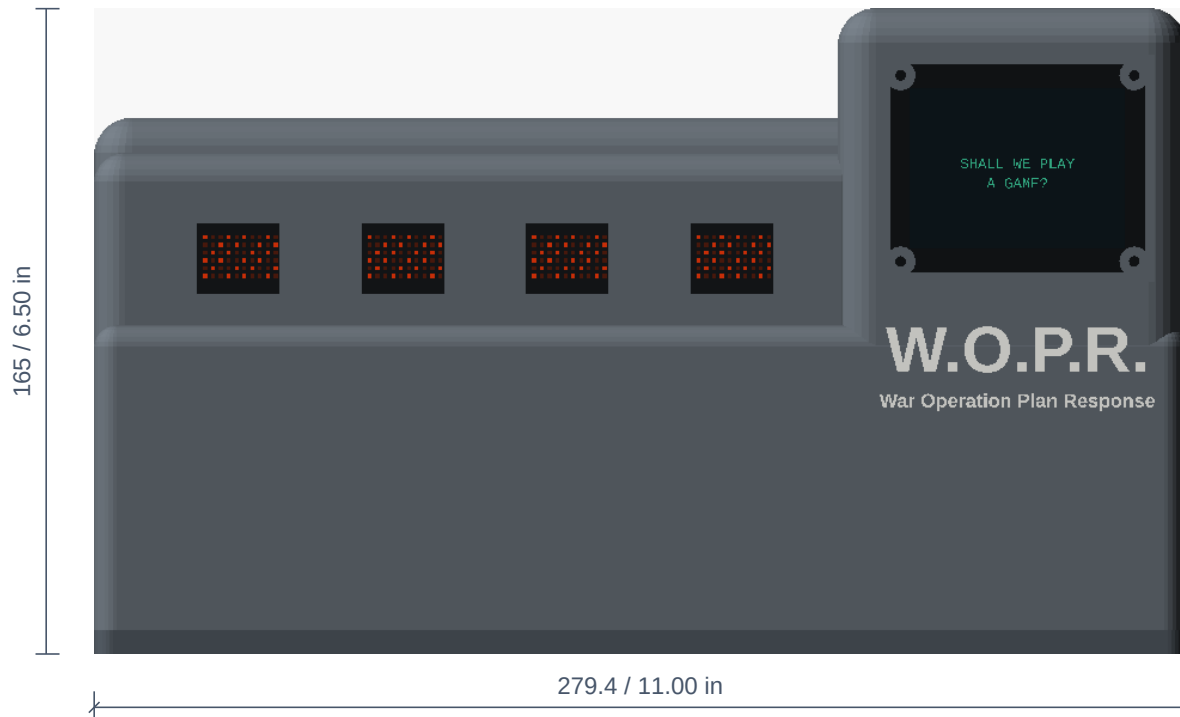
The underside opens for installation and support removal. The panel doors are decorative.

User-confirmed layout: four boards per long side, eight total. Board spacing is a first-draft choice.

Overall envelope and panel placement

All views below come from the same SCAD geometry. Dimensions are nominal, not a film-prop measurement.

PRIMARY LONG SIDE



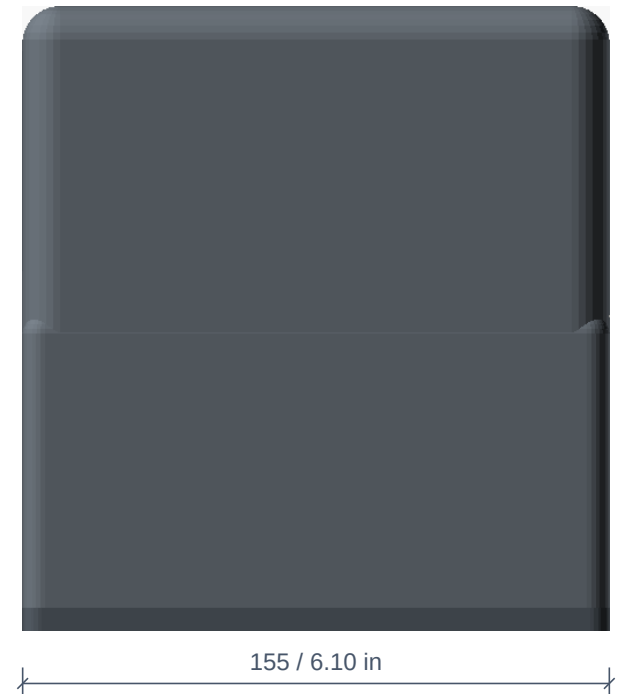
Main light centers: X = 37, 79, 121 and 163; Z = 101.

TFT center: X = 236; Z = 124. Lower case top: Z = 128.

Spine top: Z = 137. Tower top: Z = 165.

Case splits at X = 140 for a smaller printer.

TOWER END



Front display pocket:

65.6 wide x 53.6 high

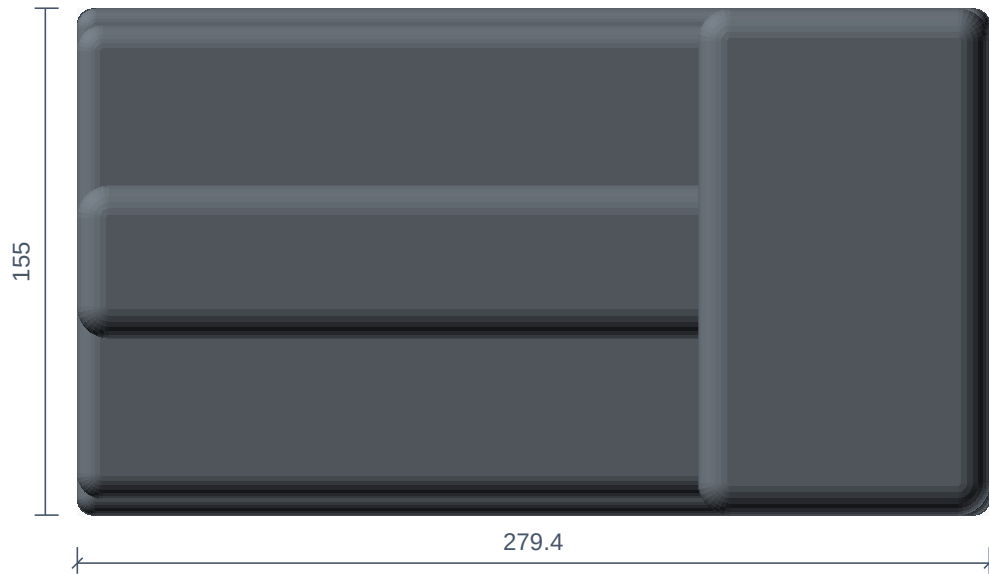
Seed pocket, each:

21.6 wide x 18.4 high

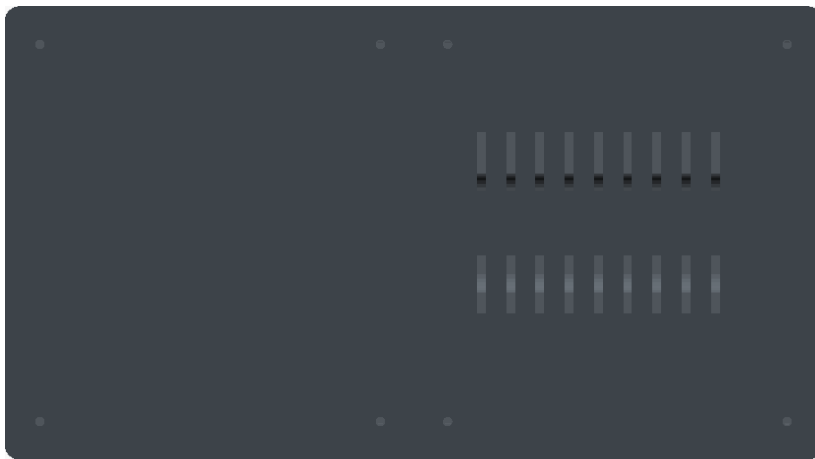
Top, rear and service access

Common axes: X along the 11-inch length, Y front to rear, Z up from the base.

TOP



OPPOSITE LONG SIDE



BOTTOM / TWO REMOVABLE HALVES

Eight M3 base screw clearances: diameter 3.3.

Shell pilot holes: diameter 2.5; fit-test first.

Bottom vent slots: 3 x 20.

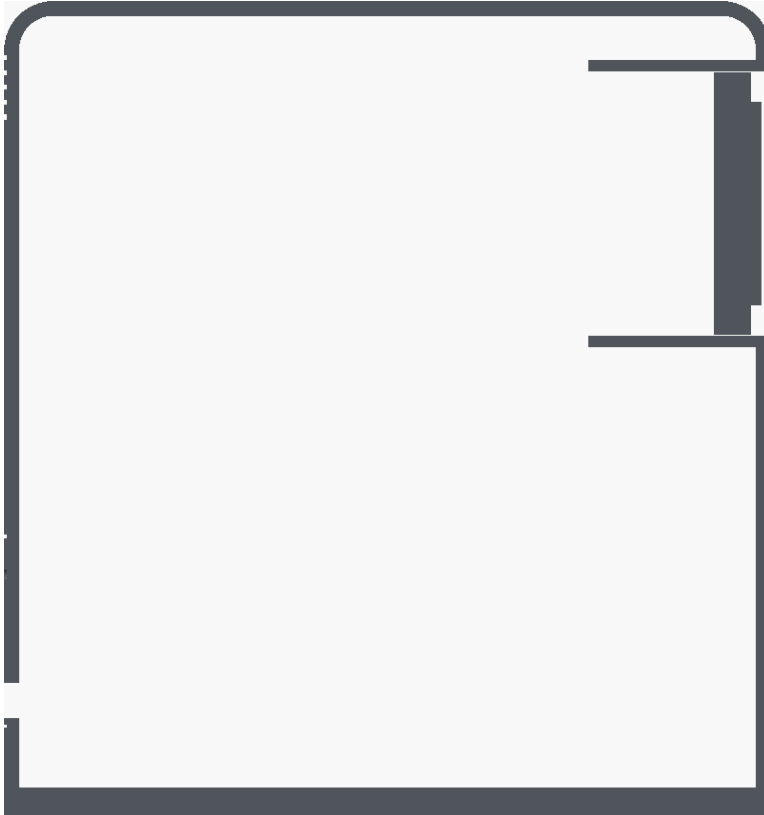
Rear cable opening: 12 x 7.

Shell and base seams are glued using splice strips.

No controller or power supply is specified yet.

A full inch behind the display

Section through X = 236. The bay is open at the rear for wires and installation from below.



SCAD SECTION / LOOKING ALONG X

DEPTH CHAIN / DETAIL



Pocket: 65.6 x 53.6, around a nominal 65 x 53 board.

Four mounting ears: diameter 2.7 clearance.

V2 hole pitch: 59.690 x 47.498 (landscape).

Front glass is nominally 2 mm below the case face.

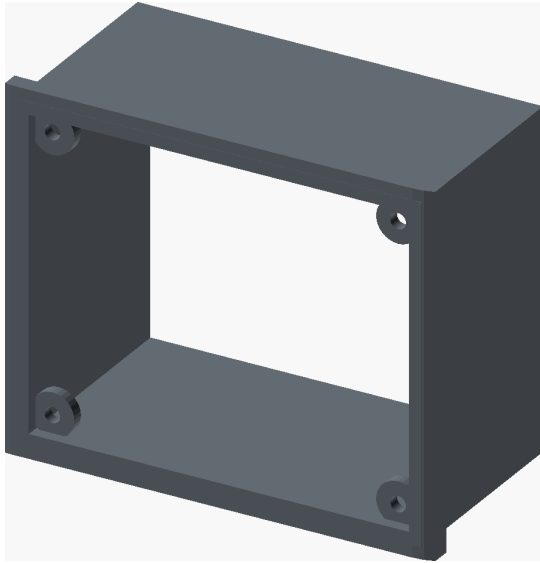
Fit-test glass-to-PCB stack and connector clearance.

The 25.4 mm bay excludes the 9.5 mm hardware depth.

Mount positions follow manufacturer CAD; assembly thickness and tolerances still need a physical check.

Gray + white, in the same print

First-draft geometry verified digitally. Physical fit and slicer toolpaths are not yet verified.



PRINT FILES

right-shell-two-color.3mf: one object, gray + white parts.

gray-left.stl / gray-right.stl: split structural case.

white-right.stl: aligned lettering; import with gray-right.

base-left.stl / base-right.stl: removable underside.

splice.stl: print four. fit-coupon.stl: test the display mount.

Nominal shell: 3 mm. White text: 0.8 mm deep, flush.

Use internal supports; remove them through the open bottom.

Start at 0.2 mm layers. Use a bed at least 180 x 180 mm.

Assign the two filament colors explicitly in the slicer.

PRINT THE DISPLAY FIT COUPON FIRST

CHECKED: Seven STL exports are watertight. Structural parts are connected solids.

CHECKED: 3MF contains two material parts. PDF has six pages with SCAD views.

NOT CHECKED: Actual print, hardware fit, touch access, electrical power or operating temperature.

Proportions are adapted from visual references. This is a review draft, not a measured replica.

SOURCES / MANUFACTURER DATA AND VISUAL REFERENCE

Seed product + board CAD

https://wiki.seeedstudio.com/rgb_matrix_for_xiao/

Adafruit #3315 / 65 x 53 x 9.5 mm

<https://www.adafruit.com/product/3315>

Adafruit V2 mounting geometry

<https://github.com/adafruit/Adafruit-2.4-TFT-FeatherWing-PCB>

Miniatua / visual proportions only

<https://www.miniatua.com/work/wopr/>